

HR Attrition Analysis

Business Intelligence Dashboard & Strategic Recommendations

Dataset:	1,470 Employees
Period:	Cross-sectional Analysis
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Agenda

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Data Model & Star Schema

FACT & Dimension tables — Power BI architecture

02

Executive KPIs

Overall attrition rate, income gap, headcount summary

03

Attrition by Dept & Role

Department, job role breakdown and comparison

04

Key Drivers of Attrition

Overtime, compensation, satisfaction, experience level

05

Outliers & Trends

Anomalies, high-risk segments, and pattern analysis

06

Strategic Recommendations

5 actionable initiatives with timelines & expected impact

Data Model — Star Schema

Power BI Architecture — 1 Fact Table + 4 Dimension Tables

DIM_Department

DepartmentID (PK)

DepartmentName

Division

CostCenter

DIM_Employee_Profile

EmployeeNumber (PK)

Gender | MaritalStatus

EducationField

BusinessTravel

FACT_Employee

EmployeeNumber (PK)

Age | MonthlyIncome

YearsAtCompany

Attrition | OverTime

Department (FK) | JobRole (FK)

DIM_JobRole

JobRoleID (PK)

JobRole

Department

Level

DIM_Satisfaction

EmployeeNumber (PK)

JobSatisfaction

EnvironmentSat.

WorkLifeBalance

All DIM tables connect to FACT_Employee via EmployeeNumber | Relationships: 1-to-1 | Cross-filter: Single direction

Executive KPI Summary

1,470 Employees | 237 Attrited | 16.12% Overall Attrition Rate

1,470

Total Employees

Full dataset analyzed

237

Attrited Employees

Left the organization

16.12%

Attrition Rate

Industry avg ~13%

\$6,503

Avg Monthly Income

Active employees

\$4,787

Avg Income (Attrited)

\$2,046 gap vs active

5.1 yrs

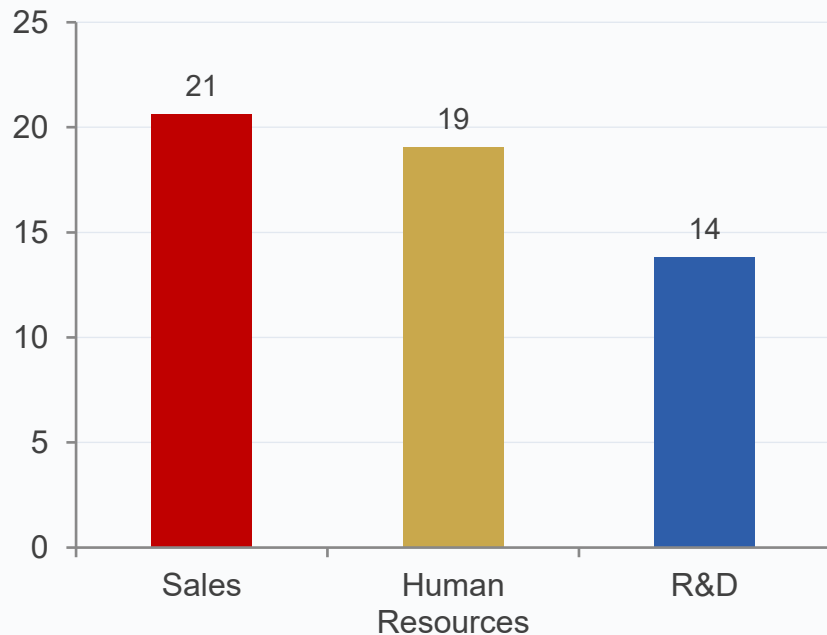
Avg Tenure (Attrited)

Leave before 6 years

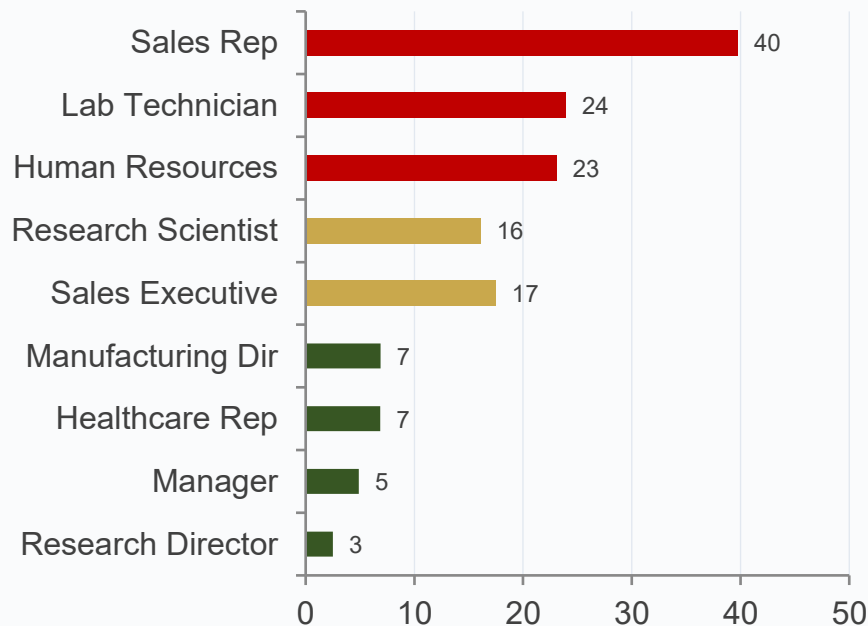
Attrition by Department & Job Role

Sales and HR departments show highest risk — Sales Representative at 39.8%

Attrition Rate by Department (%)



Attrition Rate by Job Role (%)



Key Drivers of Attrition

Overtime, compensation gap, low satisfaction & lack of equity are the top predictors

CRITICAL



Overtime

30.5%

vs 10.4%

Employees working overtime leave at 3× the rate of those who don't.

CRITICAL



Compensation Gap

\$2,046

/month gap

Attrited employees earned \$2,046 less per month than active staff.

HIGH



Job Satisfaction

22.8%

score 1 vs 11.3%

Low satisfaction (score 1) doubles attrition risk vs high satisfaction.

HIGH



Stock Options

24.4%

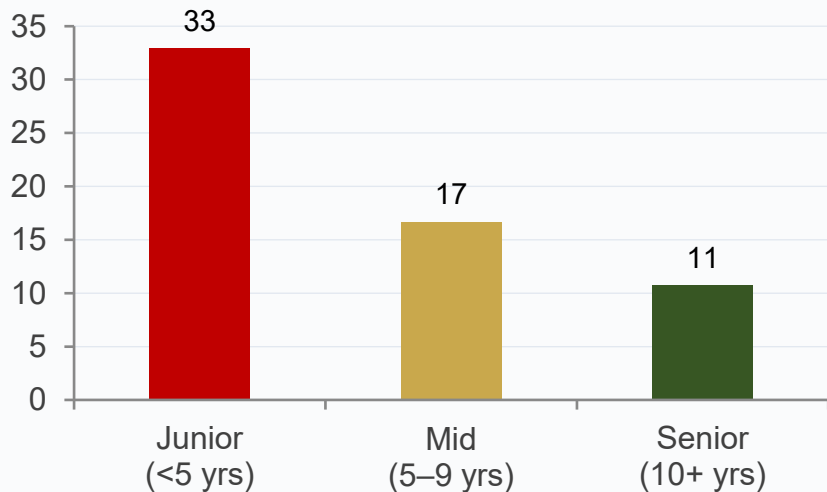
vs 9.4%

No equity (Level 0) drives 2.6× more attrition than Level 1.

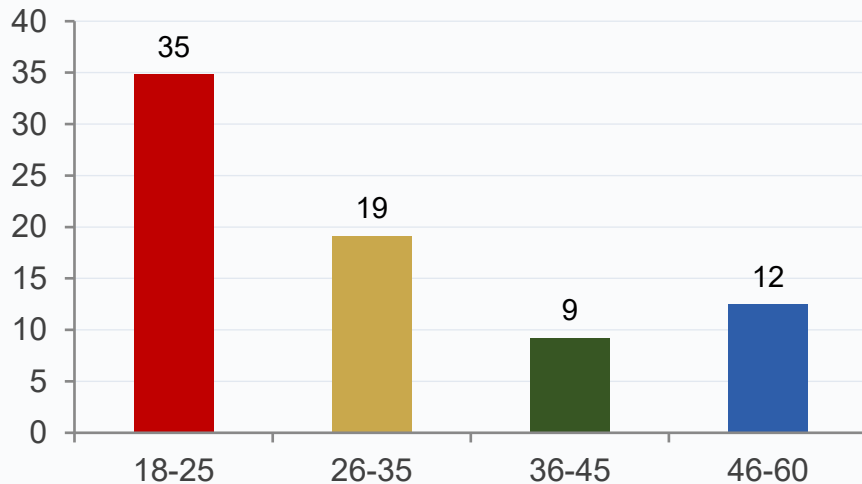
Experience Level & Demographic Analysis

Junior employees leave at 3× the rate of Senior employees — pipeline at risk

Attrition Rate by Experience Level (%)



Attrition Rate by Age Group (%)



Level	Headcount	Attrited	Rate %	Avg Income
Junior	228	75	32.9%	\$2,730
Mid	493	82	16.6%	\$4,544
Senior	749	80	10.7%	\$8,941

Outliers & Trends

Patterns that deviate significantly from the company average (16.12%)

Sales Rep Attrition

39.8%

+23.7pp | vs avg 16.1%

Highest single role — compensation-driven

Overtime Workers

30.5%

+20.1pp | vs 10.4% no OT

3× multiplier — strongest single predictor

Junior Employees <5 yrs

32.9%

+22.2pp | vs 10.7% Senior

Pipeline erosion — early career exits

No Stock Options

24.4%

+15.0pp | vs 9.4% Level 1

Equity programs cut attrition by 2.6×

Low Env. Satisfaction

25.4%

+11.9pp | vs 13.5% High

Workplace culture investment has clear ROI

Research Director

2.5%

-13.6pp | vs avg 16.1%

Lowest role — seniority & income as shield

⚠️ High-Risk Employee Segment

79 employees meet all 3 risk criteria — attrition rate: 48.1%



Overtime = Yes

Working beyond standard hours



Job Satisfaction ≤ 2

Low or very low satisfaction score



Monthly Income $< \$5,000$

Below median compensation

48.1%

Attrition Rate

among 79 high-risk employees

vs 16.12% company average

What this means:

- 38 out of 79 high-risk employees have already attrited
- 41 are currently ACTIVE — immediate intervention needed
- Cost to replace each employee $\approx$ 50–200% of annual salary
- Proactive action could retain up to 30 employees

Strategic Recommendations

5 data-driven actions to reduce attrition from 16.12% to below 10%

01

CRITICAL

Sales Rep avg income \$2,626 — lowest of all roles (39.8% attrition)

Benchmark salaries + performance bonuses + clear promotion path

 **Sales Rep attrition 39.8% → <20% | Highest ROI initiative**

 0–3 months

02

CRITICAL

Overtime drives 30.5% attrition vs 10.4% without OT (3× multiplier)

Cap OT at 10 hrs/week + mandatory rest periods + flex-time

 **~50% reduction in OT-driven exits | Affects 416 employees**

 1–2 months

03

HIGH

Employees <5 yrs tenure leave at 32.9% vs 10.7% for seniors

Mentorship program + fast-track promotions + skills development fund

 **Junior attrition 32.9% → <18% | Protects talent pipeline**

 2–4 months

04

HIGH

No stock options = 24.4% attrition vs 9.4% with Level 1 equity

Extend ESOP to all Job Level 1 employees + 2-year vesting cliff

 **2.6× retention improvement | ROI 4:1 within 2 years**

 3–6 months

05

MEDIUM

79 high-risk employees: OT + Low Satisfaction + Income <\$5K

Monthly HR review of flagged employees + proactive manager conversations

 **Early intervention retains ~30 employees | Saves \$1.2M+**

 Ongoing

Thank You

Key Takeaways

- Overall attrition rate is 16.12% — above industry average of ~13%
- Sales Representatives (39.8%) and Lab Technicians (23.9%) are highest risk roles
- Overtime is the single strongest predictor — 3× attrition multiplier
- Compensation gap of \$2,046/month between active and attrited employees
- 5 targeted initiatives can reduce attrition below 10% within 6 months

Interactive Dashboard Preview

Power BI | 3 Pages | Star Schema | 10 DAX Measures

Page 1 — Executive Overview

◆ 4 KPI Cards (1,470 | 237 | 16.12% | \$6.5K)

◆ Attrition by Dept & Job Role

◆ Satisfaction Heatmap Matrix

◆ Donut Chart — Yes/No Split

Page 2 — Key Drivers

◆ Overtime Impact (30.5% vs 10.4%)

◆ Experience & Job Level Analysis

◆ Stock Options & Satisfaction

◆ Marital Status & Business Travel

Page 3 — Outliers & Risk

◆ High Risk Count = 79 employees

◆ Income Gap = \$2,046/month

◆ Scatter: Income vs Attrition Rate

◆ Age Group Breakdown

Attrition Reduction Roadmap

Projected attrition reduction if all 5 initiatives are implemented within 6 months

Initiative	Attrition Reduction	Timeline
Overtime Management Policy	-2.5%	1–2 months
Sales Rep Compensation Revision	-2.2%	0–3 months
ESOP Expansion (Job Level 1)	-1.5%	3–6 months
Early Career Retention Program	-1.3%	2–4 months
Predictive Attrition Dashboard	-1.0%	Ongoing



Projected Outcome:

16.12%



9.62%

▼ 40% reduction | Estimated saving: \$2.1M in annual replacement costs